<!-- Extracted from https://oracc.museum.upenn.edu/doc/help/editinginatf/primer/inlinetutorial/index.html on 2026-10-16 by a stdlib HTML->Markdown converter; see raw .html for the original. -->

ATF Inline Tutorial

This page gives a tutorial on how to type graphemes and other inline content in ATF.

Graphemes

Although you will see references to GDL here, use of ATF in typing texts does not require any knowledge of GDL. GDL, the Grapheme Description Language, is the name for the model and XML vocabulary used to describe graphemes to computers. This tutorial is actually a subset of the full GDL specification [http://oracc.org/ns/gdl/1.0/index.html] intended to make it easy to start using ATF.

Characters

These days most ATF is generated in Unicode. However, it can also be restricted to ASCII characters, for which we define simple equivalents for the characters used in cuneiform transliteration which are not in the ASCII character set. The following table gives the ASCII sequences and the Unicode codepoints to which the ATF processor translates them. Certain conventions are not used in CDLI-strict notation; this is indicated in another column.

ATF Character Conventions
| Unicode-ATF | ASCII-ATF | Character | CDLI-Strict?1

|

1Characters not in the strict repertoire are not permitted in CDLI archival ATF.

| 2Lowercase `x` is permitted only in sign values; in sign names, only uppercase `X` is permitted as a notation for subscript-x. In sign names, lowercase `x` is an operator.

| sz | š | U+161 | yes

| SZ | Š | U+160 | yes

| s, | ṣ | U+1E63 | yes

| S, | Ṣ | U+1E62 | yes

| t, | ṭ | U+1E6D | yes

| T, | Ṭ | U+1E6C | yes

| s' | ś | U+015B | yes

| S' | Ś | U+015A | yes

| ' | ʾ | U+02BE | yes

| 0-9 | subscript ₀-₉ | U+2080-U+2089 | yes

| x2 | subscript ₓ | U+208A | yes

| X2 | subscript ₓ | U+208A | yes

| h, | ḫ | U+1E2B | no

| H, | Ḫ | U+1E2A | no

| j | ŋ | U+014B | no

| J | Ŋ | U+014A | no

Signs

Signs may be simple values or names, or compound sign names. Compound signs are made up of simple names combined with operators or affected by modifiers or sign-variant indicators. Names and values are one part of number signs. The forms of the various kinds of signs, as well as the compound sign operators, modifiers and variant indicators, are described in turn below.

Values

In ATF a sign value is a sequence of lowercase letters followed by optional subscript digits:

```
a      a₂    babbar      dug₄      ṣe₂    ša₁₃
```

Names

In ATF a simple sign name is a sequence of uppercase letters followed by optional subscript digits:

```
A      BA     ŠA₃    GILIM
```

A sign name in a transliteration conventionally means either that the sign is clear but its reading is uncertain, or that the sign is being used as a logogram. ATF has some simple rules to mark the difference between these two:

- You can always put a dollar sign (`$`) before a sign name to indicate that its reading is uncertain.
- You can always put a tilde (`~`) before a sign name to indicate that it is a logogram.
- You can tell the ATF processor what the meaning of the unmarked form is in an ATF protocols section.

As a result of these rules, `$AN` always means "the AN-sign is there but I am not sure which reading to choose" and `~AN` always means "the AN sign is a logogram here". The meaning of `AN` can be configured to mean either one. By default, in Sumerian language context the meaning of `AN` is equivalent to `$AN`. In all other language contexts, the meaning of `AN` is equivalent to `~AN`. This means that typing logograms in Akkadian is as easy as:

```
ša AN
```

Two special classes of sign name are signlists and numerical sign names. Numerical sign names match the pattern `N<DIGITS>`. Signlist names consist of an uppercase alphabetic prefix and an ASCII digit suffix; the prefix is the name of the sign list and the suffix is the number of the sign in that list. Prefixes fall into one of two groups. Generic signlist prefixes consist of any one or two uppercase letters followed by `SL`; hence, `CDSL`, `PSL`, `PCSL` are all valid signlist prefixes. The second group is the built-in set of historic sign lists.

In ATF signlist names and numerical sign names look like this:

```
N01       N34       LAK37        ZATU777
```

Built-in Sign List Names
| Name | Bibliography

| ABZ | R. Borger, *Assyrisch-babylonische Zeichenliste* (*AOAT* 33; Neukirchen-Vluyn 1978)

| BAU | E. Burrows, *Archaic Texts* (*UET* 2; London 1935)

| HZL | C. Ruster and E. Neu, *Hethitisches Zeichenlexikon* (Harrassowitz Verlag 1989)

| KWU | N. Schneider, *Die Keilschriftzeichen der Wirtschaftsurkunden von Ur III* (Rome 1935)

| LAK | A. Deimel, *Liste der archaischen Keilschriftzeichen* (*WVDOG* 40; Berlin 1922)

| MEA | R. Labat, *Manuel d'épigraphie akkadienne* (6th ed. Paris 1988)

| MZL | R. Borger, *Mesopotamisches Zeichenlexikon* (*AOAT* 305; Ugarit-Verlag 2003)

| REC | F. Thureau-Dangin, *Recherches sur l'origine de l'écriture cunéiforme* (Paris 1898)

| RSP | Y. Rosengarten, *Répertoire commenté des signes présargoniques sumériens de Lagash* (Paris 1967)

| ZATU | M. Green and H. J. Nissen, *Zeichenliste der Archaischen Texte aus Uruk* (*ATU* 2; Berlin 1987)

Qualified

Signs which have the special subscript `x` must be qualified in ATF by placing the sign name in parentheses immediately after the sign value:

```
baₓ(PI)          ZAX(LAK384)
```

Note: in sign values, use subscript `x`; in sign names, use uppercase `X`.

Number

In ATF a number sign conforms to the pattern:

```
REPETITION '(' GRAPHEME ')'
```

where `REPETITION` is either a number giving the repetition factor or the letter `n` or the combination `n+DIGITS` (in sign names or compound signs use `N` instead of `n`). The `GRAPHEME` is a sign value or sign name, including compound signs.

The following examples illustrate a few basic ATF numerical forms:

```
1(N₀₁)      4(ban₂)      1(aš@c)     n(geš₂)       n+1(aš)
```

The notation `n(aš)` means: some quantity in the `aš` system which is not determinable from the traces on the tablet. The notation `n+1(aš)` (where '1' could be any number) means: a quantity in the `asz` system which is damaged or lost and which is at least 1 but may be more. ATF does not use the notation `x+1(aš)`.

See also the pages on numbers, metrology and mathematical notations.

Modifier

In ATF the at-sign (`@`) precedes each modifier; multiple modifiers may be given in which case each modifier requires its own at-sign. The entire sequence of modifiers (and allographs, described below) belongs to the immediately preceding sign or group. Sign names and values with modifiers and/or allographs following them should not be be treated as compounds.

ATF Modifier Overview
| Modifier | ATF | Example | Sign

curved	@c	AŠ@c
flat	@f	1(N01@f)
gunu (4 extra wedges)	@g	DU@g
sheshig (added še-sign)	@s	DU@s
tenu (slanting)	@t	GAN$_2$@t
nutillu (unfinished)	@n	SAG@n
zidatenu (slanting right)	@z	AŠ@z
kabatenu (slanting left)	@k	AŠ@k
vertically reflected	@r	U@r
horizontally reflected	@h	N07~a@h
rotated	@<DIGITS>	NAGA@180
variant	@v	4(ban$_2$)@v

Modifiers on numerical graphemes may go inside or outside the closing parenthesis depending on the naming schema for values and sign names used by the style manual or sign list for an individual project.

Allograph

In ATF an allograph, or systemic sign variant, is introduced by the tilde-prefix (`~`); the sequence of characters following the tilde is restricted to ASCII digits and lowercase letters, **except for x**.

```
|EN~a|  |EN~b|  |GA₂~a1|  |GA₂~a2|  |GEŠTU~axŠE~a@t|
```

The use of `x` as an operator in examples like the last one in the line above is the reason for excluding `x` from the characters allowed in allograph sequences.

The special allograph `~v` is used instead of `~x` to indicate that the form is some variant of the sign in question but the specific variant is not identified.

The special allograph `~t` is used to indicate tokens, e.g., `1(N₀₁~t)`.

Note that the allograph mechanism is not the way that unusual sign forms are notated in ATF; for this the normal exclamation mark (`!`) is used. The allograph mechanism is provided to support systematic subdivision of sign-forms relative to extant sign lists or sign name descriptions.

Formvars

Form variants are preceded by the backslash character (`\`) and consist of lowercase letters and or digits.

Compound

The difference between a simple sign and a compound sign is that a a compound sign is a sequence of sign names which contains at least one operator, i.e., a character which represents a relationship between multiple graphemes. In ATF the set of characters used for operators is: `×%@&.:+`.

In ATF compound graphemes are enclosed at the outer level in vertical bars ("pipes", `|...|`):

```
|KA×A|
```

Signs are frequently modified or operated on as a group; parentheses are used to group multi-part constituents:

```
|GA₂×(ME.EN)|      |(GI&GI)×ŠE₃|
```

Note that modifiers and allographs must not be placed after the closing pipe; instead, they must be put inside the pipe adding grouping characters if necessary:

```
|GA₂~a×EN|      |GA₂×EN~a|          |(GA₂×EN)~a|
```

Th examples above all mean different things. The first, `|GA₂~a×EN|`, means: "the a-allograph of the sign GA_2 containing sign EN". The second, `|GA₂×EN~a|`, means: "GA_2 containing the a-allograph of sign EN". The third, `|(GA₂×EN)~a|`, means: "the a-allograph of the group consisting of sign GA_2 containing sign EN". In example three the bad form `*|GA₂×EN|~a` would result in a parse error.

Each of the compound operations has its own ATF notation as summarized in the table below:

Summary of Compound Grapheme Operators in ATF/GDL
| GDL | ATF | Example | Sign

beside	.		DU.DU	
joining	+		LAGAB+LAGAB	
containing	×		GA_2×AN	
containing/group	×		GA_2×(ME.EN)	
above	&		DU&DU	
crossing	%		GI%GI	
opposing	@		LU_2@LU_2	
repeated	3×		3×AN	
repeated	4×		4xLU2	

Punctuation

Several types of cuneiform punctuation are supported in ATF and all of them must be preceded and followed by a space (in the case of `*` and `/` the punctuation may be immediately followed by a sign name in parentheses and then the following space). The recognized punctuation codes are:

*** = Bullet**
: The "1" used at the start of each line in lexical texts, omen compendia, etc..

***(GRAPHEME)**
: Generic punctuation; most often used where scribes use signs other than a "1" at the start of the line in lexical texts, but may be used to transliterate arbitrary or unusual kinds of punctuation

that are not otherwise covered below.

: = .
:

The vertical "colon" sign often found in commentaries.

N.B.: If the single colon occurs within a word it must be transliterated with the grapheme name form `P2`

**:' (colon+right-quote) = **
: Borger MZL 592 variant b; a variant on the vertical two-wedge colon

**:" (colon+double-quote) = **
: The diagonal "colon" sign often found in commentaries. Note that the three different double-wedge colon signs are mnemonically two-dots, two-dots-prime and two-dots-double-prime

**:. = **
: The triple-wedge "colon" sign sometimes found in commentaries.

:: = ??
: (A colon convention defined in the SAA style manual, form unspecified.)

**/ = **
: Word divider; if unqualified, this is the single vertical wedge word-divider as used, e.g., in Old Assyrian texts. May be qualified as, e.g., `/(P2)`.

Punctuation Sign Names

The punctuation signs may also be transliterated using the following names: P1 (); P2 (); P3 (); P4 (); MZL592~b (as `:'`).

Metadata

In ATF there are several general ways of specifying information about graphemes:

Flags
: Flags are single characters which are placed immediately after a grapheme; they may occur in any order. The flag characters in ATF are: `!?*#`. Flags may appear only immediately after a grapheme (after the parenthetic part of a qualified grapheme), and are not permitted within compound signs. Flags may be given after the closing pipe of a compound sign.

Brackets
: Brackets are matching pairs of single or double characters. The matched pairs in ATF are (the ellipsis in the following is a conventional indicator for text between the brackets): `(...)` `[...]` `{...}`

`{(...)} <...> <<...>> <(...)>`. Brackets are not permitted within the body of graphemes or within compound signs.

Shifts
: Shifts are combinations of a percent-sign (`%`) and a label which alter the current value of a property. The value remains current until another shifter for the same property is encountered; or until the closing of the nearest enclosing bracket; or until the end of the line. Sample shifts include: `%s %akk %1`. A table of all of the shifts, properties and values is given later on in this tutorial. Shifts are not permitted within compound signs. Shifts must always be followed by one or more spaces.

Breakage

Partially broken signs are flagged by putting a hash (`#`) after the grapheme. Signs which are completely missing from the tablet are enclosed in square brackets.

```
ba#        [a]-ba      mud×(|ZI&ZI.A|)#
```

Other flags

Collation, uncertainty and remarkability are flagged by `*`, `?` and `!` respectively. If a grapheme is flagged as remarkable it may indicate a correction or an unusual form. Corrections are often followed by the actual sign in parenthesis, and this convention is supported but not required in ATF transliterations.

```
a*                ki?             šum!               ki!?*(DI)
```

Glosses

ATF divides glosses into three types:

Determinatives
: Determinatives include semantic and phonetic modifiers, which may be single graphemes or several hyphenated graphemes, which are part of the current word. Determinatives are enclosed in single brackets `{...}`; semantic determinatives require no special marking, but phonetic glosses and determinatives should be indicated by adding a plus sign (`+`) immediately after the opening

brace, e.g., `AN{+e}`. Multiple separate determinatives must be enclosed in their own brackets, but a single determinative may consist of more than one sign (as is the case with Early Dynastic pronunciation glosses).

Linguistic
: Linguistic glosses are defined for the purposes of this specification as glosses which give an alternative to the word(s) in question. Such alternatives are typically either variants or translations. Linguistic glosses are enclosed in the double brackets `{{...}}`.

Document-oriented
: Document-oriented glosses are used for scribal comments on the document including 10-marks, line-count summaries and asides such as *he-pi₂* ("(text) broken"). Document-oriented glosses are enclosed in the compound brackets `{(...)}`.

Glosses must have a space or hyphen on one side or the other. They may have spaces on both sides. Glosses may not touch directly both the preceding and following graphemes; nor may they have hyphens at both ends.

```
{d}utu   larsa{ki}   {+u3-mu2}u2-mu11    AN{+e}

du3-am3{{mu-un-<(du3)>}}

{(1(u))}    {(%a he-pi2 eš-šu2)}
```

Presence

The status of one or more graphemes' presence/absence often requires notation. The following bracketings support the common practices in Assyriological transliteration:

(...)
: The enclosed graphemes may be present but this is not certain; normally used within [...] as in

```
[x (x) x]
```

<...>
: The graphemes must be supplied for the sense but are not present

```
mu-un-<pa3>-da
```

<<...>>
: The graphemes are present but must be excised for the sense

```
mu-un-<<an>>-pa₃-da
```

<(...)>
:

1. **[AFTER BOUNDARY]** The graphemes are implied because the scribe has left a blank space on the tablet; common in liturgies and some types of administrative texts. `BOUNDARY` can be space, hyphen, etc.
2. **[AFTER GRAPHEME]** The graphemes are the text meant by a surrogate grapheme such as `MIN`; common in lexical texts.

```
1. {d}suen he₂-me-en
2. {d}nanna <(he₂-me-en)>

1. a    = %a mu-u₂
2. illu = %a MIN<(mu-u₂)>
```

Surrogates are defined in the XTF2 schema because their content model is `l.inner`.

Note: in all of these cases except the last there must be a space or hyphen before the opening bracket and after the closing bracket.

Scripts

A simple mechanism for identifying distinct scripts on a single document is provided by the percent-digit commands which consist of a percent sign followed by a single digit: `%0 %1 %2 %3 %4 %5 %6 %7 %8 %9`. The characteristics of the scripts identified by numbers can be specified in the protocols section at the start of an ATF file.

By default, the normal sized, normal form script is `%0`; this only needs to be specified rarely. By default, the smaller script used for glosses is `%1`. Two other default scripts are `%2` = Assyrian and `%3` = Babylonian as a contrastive pair in neo-Assyrian documents.

This example shows how to enter a gloss which is in smaller script:

```
mu-un-šum₂{%1 szu}
```

Languages

Defining the default language

At the start of the ATF file, amongst the other protocols [http://oracc.museum.upenn.edu/ns/xtf/1.0/atftut.html#Protocols], you need to define the language of your (ancient) text. (For instructions on how to define the language of your translation, see the page on translations [http://oracc.museum.upenn.edu/doc/builder/translation].)

After the &-line but before the text begins, enter a single protocol line which begins `#atf: lang`, followed by a space and the relevant language or dialect code in lower-case. This example is for Neo-Babylonian:

```
#atf: lang akk-x-neobab
```

This line ensures that all transliterated and lemmatised words in the text will be treated as Neo-Babylonian--unless you explicitly mark otherwise, as described below.

Shifting to other languages

To shift to a different language within the text, write a percent sign followed immediately by the relevant language code. You will also need to explicitly signal the shift back to the default language. For instance, if you had not defined Emesal as the default alternative language in the first example you could write:

```
8. %e še-eb %s e₂-kur-ra ba-du₃-a-bi
```

As before, the text is assumed to switch back to the default language at the start of every new line.

Language codes

Here is a list of the most frequently used language and dialect codes. The full set, including peripheral dialects of Akkadian, is given on the Language Tags [http://oracc.museum.upenn.edu/doc/developer/l2/langtags/] page of the developer documentation.

The Main Language and Dialect Codes
| Language or dialect | Protocol Code | Inline Code(s) | Notes

Akkadian	(none: must specify dialect too)	a *or* akk

| Early Akkadian | akk-x-earakk | eakk | For pre-Sargonic Akkadian.

Old Akkadian	akk-x-oldakk	oakk

Ur III Akkadian	ua	ur3akk

Old Assyrian	akk-x-oldass	oa

Old Babylonian	akk-x-oldbab	ob

Old Babylonian peripheral	akk-x-obperi	

Middle Assyrian	akk-x-midass	ma

Middle Babylonian	akk-x-midbab	mb

Middle Babylonian peripheral	akk-x-mbperi	

Neo-Assyrian	akk-x-neoass	na

Neo-Babylonian	akk-x-neobab	nb

Late Babylonian	akk-x-ltebab	nb

Standard Babylonian	akk-x-stdbab	sb

| Conventional Akkadian | akk-x-conakk | ca | The artificial form of Akkadian used in lemmatisation Citation Forms.

| normalised | (none: main text must be transliteration) | n | Used in lexical lists and restorations; try to avoid wherever possible.

| transliterated (graphemic) Akkadian | (none: must specify dialect too) | g | Only for use when switching from normalised Akkadian.

Hittite	hit	h *or* hit

| Sumerian | sux or sux-x-emegir | s, sux, *or* eg | The abbreviation `eg` stands for Emegir (main-dialect Sumerian).

Emesal	sux-x-emesal	e, es

Syllabic	sux-x-syllabic	sy

Udgalnun	sux-x-udgalnun	u

Proximity

A general facility for annotating graphemic proximity is provided with the notation `$<zone>`, where `zone` is an arbitrary region of the surface defined only by the transliterator and represented by a single digit. Search engines may provide facilities to find multiple graphemes with the same zone code (in the same line) and possibly to relate grapheme proximity to the difference between zone codes (i.e., in `a$1 e$2 i$3`, the `i` grapheme may be considered closer to `e` than to `a` (the ordering relationships of zone codes are likely to be problematic, however). This is an experimental feature intended for use in exploring the graphotactics of Early Dynastic texts.

Intrusions

To indicate that there is any kind of newline within a transliterated sequence of words or graphemes that is in the same ruled case (usually, but not always, indented), use the semi-colon (`;`).

To indicate that there is a line-break in the transliterated line (i.e., normally one which is separately ruled), use a double slash sequence (`//`).

To indicate that an unknown number of signs is missing, use an ellipsis (`...`).

If it is necessary to indicate any other information which is not part of the grapheme sequence the compound brackets `(#...#)` may be used. This feature should be used rarely if ever.

```
mu-un;-e₃          [...]          [(...)]
```

Words

For the purposes of transliteration, a "word" is anything between spaces, including isolated and uninterpretable signs.

In ATF words are separated by spaces, and graphemes within words are joined by hyphens. Note that periods (`.`) are only permitted inside compound graphemes.

Simple choices in the transliteration of single graphemes may be expressed by separating the graphemes with a slash (`/`). More than one choice may be given, but each sequence of choices only applies to a single grapheme.

The fact that signs are inscribed on the object in a different order than they are transliterated may be indicated by joining graphemes with the colon (`:`) instead of the hyphen. This mechanism is a convenient shorthand which is intended to cover cases of occasional reversal in

sign order. It is only available within words, it is not permitted in compound signs, and it may only be used with pairs of reversed graphemes. A different, completely general, mechanism for indicating more complex reorderings is provided under the concept of Multiplexing and is explained in the description of document structure.

```
a-ba mu-un-ba-al-e     KI/DI-bi     LAGAB-DUL₃      mu:un-du₃
```

18 Dec 2019 osc at oracc dot org

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Steve Tinney, 'ATF Inline Tutorial', *Oracc: The Open Richly Annotated Cuneiform Corpus*, Oracc, 2019 [http://oracc.museum.upenn.edu/doc/help/editinginatf/primer/inlinetutorial/]

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